CHEMICAL BONDING

↳ Force of attraction present between any chemical species like, atoms, ions & molecules.

Bond formation → Exothermic

Bond dissociation → Endothermic

Chemical bonds on the basis of Bond Energy

Strong bond	Weak bond
① Ionic bond	① Hydrogen bond
② Covalent bond	② Vander waal's bond
③ Co-ordinate bond	
④ Metallic bond	

Cause of chemical combination

Modern concept :- Every atom has a tendency to form Max^m. possible bonds in order to minimize its potential energy & gain stability.

Classical concept (octet rule) :-
↳ Every atom has a tendency to acheive stable outer octet (8e⁻ in valence shell).

Different theories to explain bonding

① Octet rule

② VSEPR (Valence shell electron pair repulsion theory)

③ VBT (Valence bond theory)

④ MOT (Molecular orbital theory)

Kossel-lewis Approach (octet rule):

- Based on inertness of noble gas
- Atoms can be combine either by transfer or by sharing of valence electrons in order to have an octet (8e⁻ in their valence shells).
- In case of H & He duplet ($2e^-$ in valence shell) will be completed.

How to write Lewis dot structure & bond formation of Molecules.

$$\text{No. of Bonds} = \frac{\underset{(\text{No. of atoms} \times 8)}{\text{Req. valence } e^-} - \text{Available val. } e^-}{2}$$

E.g → $\boxed{CO}$

$$\text{No. of bonds b/w C \& O} = \frac{(8\times2)-(4+6)}{2} = 3$$

$\boxed{O_3}$

$$\text{No. of bonds} = \frac{(8\times3)-(6\times3)}{2} = 3$$

$\boxed{N_3^-}$

$$\text{No. of bonds} = \frac{8\times3-(5\times3)+(1)}{2} = 4.$$

:N=N=N:

N≡N—N:

Limitations of octet rule

(i) Existence of Hypervalent compound.
$\boxed{\text{val } e^- > 8}$ E.g → PCl_5, SF_6, H_2SO_4.

(ii) Existence of Hypovalent compound.
$\boxed{\text{val } e^- < 8}$ E.g → $LiCl$, BF_3, $BeCl_2$.

(iii) Odd electron species donot follow octet rule. Eg → NO, NO_2

(iv) Existence of compounds of inert gases with oxygen & fluorine like XeF_2, KrF_2, $XeOF_2$ etc.

(v) does not explain the relative stability of the molecules being totally silent about the energy of a molecule.

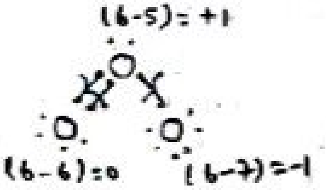
(vi) This theory does not account for the shape of molecules.

⇒ Formal charge :-

F.C on atom = Jitne hone chahiye (val e⁻ in free state) − Jitne hain (val e⁻ in given structure)

$$\boxed{F.C = \text{Total val. } e^- - [2 \times L.P + \text{Bond pair}]}$$

Eg → ① O_3



⇒ Stability of same molecules of diff str

① More stable due to symmetrical charge distribution.

② More stable due to absence of charge.

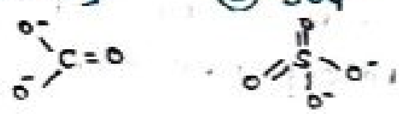
③ More stable due to -ve charge present on high EN element.

Avg. formal charge & Avg. X-O bond order in oxo-Anions:-

Avg. FC = $\dfrac{\text{total charge present on O-atom}}{\text{No. of O-atom where -ve charge can be placed}}$

Avg. X-O bond order = $\dfrac{\text{total no. of X-O bonds}}{\text{no. of X-O pairs participate in resonance}}$

① CO_3^{2-} ② SO_4^{2-}



	① CO_3^{2-}	② SO_4^{2-}
F.C on 'O' →	-2/3	-2/4
X-O bond order →	4/3	6/4

Trick	FC
=O →	0
-O →	-1
≡O or =O- →	+1

Covalent bond

↳ Formed by the equal sharing of e^-.

★ Covalency:- Total no. of covalent bond (σ & π) formed by an atom in a molecule.

VALENCY (combining capacity)

① Covalency → (Total no. of σ & π bonds)

② Electrovalency → Combining capacity in ionic comp. Eg → Na^+Cl^- → $Na^+ = +1$, $Cl^- = -1$

NOTE:-

① Max covalency of 2nd period element can not be more than '4' due to absence of d-orbitals.

② 3rd period elements can extend their covalency beyond '4' by using vacant 3d orbitals.

③ 'F' can show only 1 covalency.

Coordinate bond

↳ Formed by the unequal sharing of e^-.

↳ Formed b/w e^- donor & e^- acceptor.

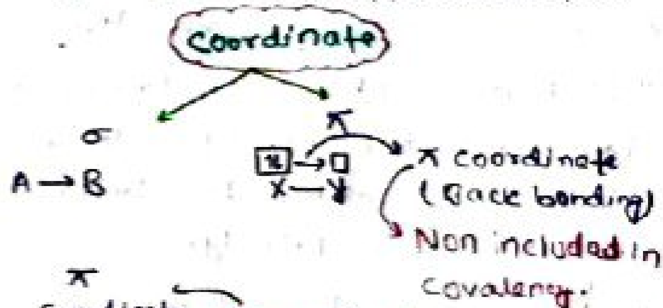
$A \rightarrow B$ σ Coordinate

or

$\overset{+}{A} - \overset{-}{B}$

↳ e^- donor = Lewis base

↳ e^- acceptor = Lewis Acid



π coordinate $C \equiv O$ or $\bar{C} \equiv \overset{+}{O}$

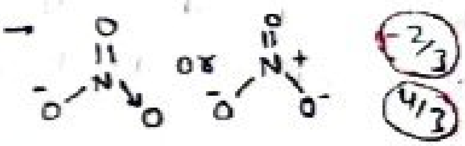
Species in which σ coordinate bond (dative bond) is present.

① O_3 → $O=O \rightarrow O$ or $O=\overset{+1}{O}-O^{-1}$

② NH_4^+ → $[\ddot{N}H_3 \rightarrow \square H^+]$

③ H_3O^+ → $[H_2\ddot{O} \rightarrow \square H^+]$

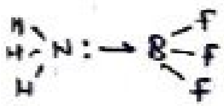
④ NO_3^- →



⑤ PCl_6^- → $(Cl^- \rightarrow PCl_5)$

⑥ BF_4^- → $F^- \rightarrow BF_3$

⑦ NH_3BF_3



⑧ PH_4^+ $[H_3P: \rightarrow H^+]$

⑨ $B(OH)_4^-$ → $B(OH)_3 \leftarrow \ddot{O}H^-$

VSEPR (Valence shell electron pair Repulsion theory):-

- Based on the repulsive interactions of the electron pairs in the valence shell of the atoms.
- Overall geometry / shape of molecule depend on 'σ' bond pairs & lone pairs.

linear — Bent

Order of Repulsion

$\boxed{l.p - l.p > l.p - B.P > B.P - B.P}$

Types of Monocentric Molecule

σ-bond, Lone pair

central Atom — $\boxed{A\ B_n\ L_m}$ — no. of L.P

no. of B.P

Monocentric Molecules		Shape / Geometry
$\boxed{AB_2L_0}$	→	Linear
$\boxed{AB_3L_0}$	→	Trigonal planar
AB_2L_1	→	Bent / V-shape
$\boxed{AB_4L_0}$	→	Tetrahedral
AB_3L_1	→	Trigonal Pyramidal
AB_2L_2	→	Bent / V-shape / Angular
$\boxed{AB_5L_0}$	→	Trigonal Bipyramidal
AB_4L_1	→	See-Saw
AB_3L_2	→	Bent 'T'-shape
AB_2L_3	→	Linear
$\boxed{AB_6L_0}$	→	Square Bipyramidal or Octahedral
AB_5L_1	→	Square pyramidal
AB_4L_2	→	Square planar
$\boxed{AB_7L_0}$	→	Pentagonal Bipyramidal
AB_6L_1	→	Distorted octahedral
AB_5L_2	→	Pentagonal planar

Telegram:- iitjeeneetnotess

⇒ How to find Hybridisation

(L.P+σ) (m+n)		Type of Hybridisation
2	→	SP (S+P)
3	→	sp^2 (S+P+P)
4	→	sp^3 (S+P+P+P)
5	→	sp^3d (S+P+P+P+d) etc.

● Trick to find n & m in AB_nL_m:-

$\boxed{\frac{\text{Total valence } e^-}{8}}$ Quotient = n → no. of σ-bond; $\frac{\text{Remainder}}{2}$ = m → no. of l.p

① BF_3 $\frac{3+7+7+7}{8} = \frac{24}{8}$ Q = 3 (n), R = 0

AB_3L_0, SP^2, Trigonal planar

② NO_2^+ $\frac{5+6+6-1}{8} = \frac{16}{8}$, Q = 2, R = 0

AB_2L_0, linear, SP

③ NO_3^- $\frac{5+(6\times3)+1}{8} = \frac{24}{8}$ = Q = 3, R = 0

AB_3L_0, SP^2, T. planar

④ SO_2 $\frac{6+6+6}{8} = \frac{18}{8}$ Q = 2 (n), R = 2, R/2 = 1 (m)

AB_2L_1, Bent, Sp^2

NOTE:- Do not apply this trick for odd electron species & species having hydrogen as S.A

⇒ If H & Halogens are S.A (Monovalent)
↳ Jitne hai utne C.A ke valence e^- kam kar do!!, Bache huye ke l.p bna do.

① CH_4

valence e^- of C = 4
no. of H = 4
l.p = 4 − 4 = 0

AB_4L_0, sp^3 tetrahedral

② NH_4^+

Valence e^- of N = 5
= (5−1) = 4
(∵ '+' hai)

AB_4L_0

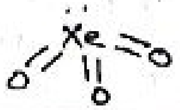
③ NH_2^-

→ (5+1) = 6
6 − 2 = 4 = 2 l.p

AB_2L_2

NOTE:- Jab compound me koi bhi charge nahi Ho aur S.A oxygen hoto sare "O" double bonded consider honge.

Eg → XeO_3



Hybridisation

↳ Mixing of orbitals

↳ Jitne milenge, utne Banenge
(A.O) (H.O)

↳ H.O are equivalent in shape & Energy.

↳ Hybrid orbital ke 2 kaam
↓ [no. of H.O = σ + l.p]
σ Banayega & l.p Rakhega

↳ Size of H.O → $SP < SP^2 < SP^3$
% P ↑ Size ↑

↳ 'S' Angle ki dukaan hai.
'P' direction ki.

⇒ Mixing hogi to

↳ S ka directional ↑se

↳ P ka angular Nature ↑se

Bond Angle :- Gen. 1 LP ↓se BA by 2.5° to 2.8°.

↳ L.P Lagao, Angle ghatao.

→ SP > SP^2 > SP^3

%S = 50% %S = 33% %S = 25%

θ = 180° θ = 120° θ = 109.5°

If Hybridisation same

If L.P is at C.A

↳ l.p ↑, B.A ↓

↓

If hybrid & L.P both are same

→ Size of S.A ↑, B.A ↑

→ Size of C.A ↑, B.A ↓

If l.p is absent at C.A

↳ give us answer A/q to hybridisation, if hybr. same, Angle same

$BF_3 = BCl_3$

AB_3L_0 AB_3L_0

SP^2 SP^2

Types of d-orbitals used in hybridis.

$SP^3d \rightarrow dz^2$

$SP^3d^2 \rightarrow dz^2, dx^2-y^2$

$SP^3d^3 \rightarrow dz^2, dx^2-y^2, dxy$

Eg → SF_6 → AB_6L_0, SP^3d^2

$S + P_x + P_y + P_z + dz^2, dx^2-y^2$

Eg → XeF_2 → AB_2L_3, SP^3d

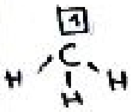
$S + P_x + P_y + P_z + dz^2$

NOTE:- Orbitals remains pure (not used in hybridisation).

Eg → (dyz, dxz)

Hybridisation In Odd e^- Species

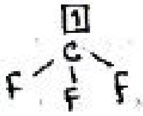
↳ Jab S.A 'H' Hoga, hybridisation me odd e^- vala dabba participate nahi karega. Eg → CH_3



∵ EN of S.A < E.N of C.A.
↳ SP^2

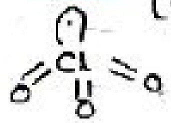
↳ Orbital having odd e^- can participate in hybridisation only when S.A is high EN.

Eg → CF_3



SP^3

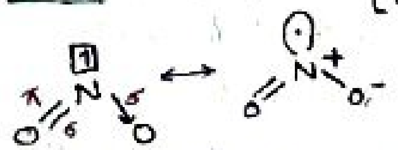
ClO_3 [$EN_O > EN_{Cl}$]



SP^3

⇒ NO_2 (V. Imp)

[$E.N_{S.A} > EN_{C.A}$]



↳ Hybridisation of N = SP^2

↳ Shape - Bent

↳ O–N–O bond Angle 120° < θ < 180°
θ ≃ 134°

↳ ~~Unpaired~~ 1σ coordinate bond is present.

↳ Acidic oxide

↳ can cause Acid rain

↳ Brown coloured gas, Paramagnetic

↳ can demerise into N_2O_4 in High P & Low T.

Hybridisation of some molecules in solid state :- Exist in form of ions.

↳ $PCl_4^+ | PCl_6^-$ — AB_4L_0 tetra sp^3 — AB_6L_0 octahedral sp^3d^2

* PBr_6^- ↳ donot exist due to large size of Br atom (steric hindrance)

PBr_4^+ ✓

Existing & Non-existing Molecules

(A) Non-contraction of d-orbitals

Existing	Non-existing
XeF_6 (sp^3d^3)	XeH_6
XeF_2 (sp^3d)	XeH_2
SF_6 (sp^3d^2)	SH_6

↳ When S.A is high EN, Zeff of C.A inc.

↳ If C.A have to use its d-orbitals in hybridisation they must be sufficiently contracted it is only possible when S.A is high EN.

(B) Improper covalency

Existing		Non-existing
NCl_3	2nd period elm	NCl_5
BF_4^-	Max. covalency = 4	BF_6^{3-}
ClF_3	F = only one	FCl_3

(C) Steric Hindrance

Existing	Non-existing
PCl_6^-	PBr_6^-
SiF_6^{2-}	$SiCl_6^{2-}$
IF_7	ICl_7 / IBr_7

Types of overlapping

↳ Always $\sigma > \pi$ (strength)

∵ Co-axial overlapping > collateral

- Co-axial / Head-on overlapping

↳ Along with Internuclear axis (INA)

↳ σ Bond formation

- Collateral / Sideways overlapping

↳ Perpendicular to INA

↳ π Bond formation

① H.O + H.O (C.A) (S.A)
H.O + P.O (C.A) (S.A) → Always σ

② P.O + P.O → σ/π

Types of P.O + P.O overlapping (σ/π)

① s–s ② s–p ③ p–p
④ p–d ⑤ d–d

① s–s — (INA) → s kevi π-bond nhi bna skta h.

s, s: x σ, y σ, z σ

② s–p — (x) σ → common par σ, baaki par zero.

s, p_x: x σ, y 0, z 0

③ p–p → common par σ, baki par π.

$p_x + p_x$: x σ, y π, z π

NOTE → Both 'p' must be same.

$p_x + p_y$: x 0, y 0, z 0

④ p–d → Uncommon par π, baki par zero.

$d_{xy} + p_x$: x 0, y π, z 0

$p_x + d_{yz}$: x 0, y 0, z 0

⑤ d–d → common par π, baaki par δ

d_{xy}, d_{xy}: x π, y π, z δ

NOTE → Both 'd' must be same.

d_{xz}, d_{yz}: x 0, y 0, z 0

Overlapping of pure Atomic orbitals

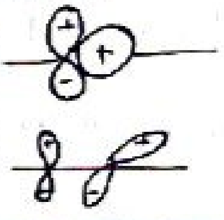
① Positive overlap

(+) + (+), (−) + (−)

② Negative overlap

(+) + (−)

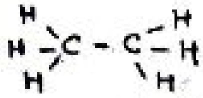
③ zero overlap



Bond Identity

① σ-Bond Indentity

ⓐ H–H (s–s) ⓑ H–Cl (s–p)

ⓒ Cl–Cl (p–p) ⓓ 

C–C (sp^3–sp^3)

C–H (sp^3–1s)

② π-bond Identity (imp)

NOTE → If C.A belongs to 2^{nd} period then there is no 'd' available, so no dπ–dπ bonds.

ⓐ SP → All are pπ–pπ

ⓑ SP^2 → first one is pπ–pπ, Rest are dπ–dπ

ⓒ SP^3 or any other → All are dπ–dπ.

Bond strength / Bond Energy

(i) Bond strength $\propto \frac{1}{\text{Size (Shell no)}}$

1s–1s > 2s–2s > 3s–3s

(ii) If size same :-

(a) s–s < s–p < p–p (for co-axial overlapping)

2s–2s < 2s – 2p < 2p–2p

$\boxed{\text{B.S} \propto \text{directional nature}}$

(b) pπ–pπ < pπ–dπ < dπ–dπ (for collateral overlapping)

3pπ–3pπ < 3pπ – 3dπ < 3dπ–3dπ

$\boxed{\text{B.S} \propto \text{extend of overlapping}}$

(iii) Lone pair-lone pare repulsion can show exceptions only in 2^{nd} period elements. (only in single-bonded atom)

$\boxed{\text{L.P–L.P repulsion} \downarrow \text{ses B.S}}$

⇒ Compare Bond strength in following

- 1s–2s < 1s–2p (directional nature of p ↑ B.S ↑)
- p–p (co-axial) > p–p (collateral)
  σ π
- N–N < P–P
- O–O < S–S } → (L.P–LP repulsion)
- $Cl_2 > Br_2 > F_2 > I_2$
- 1s–1s > 2p–2p > 2s–2p > 2s–2s > 3s–3s
- 3pπ–3pπ < 3pπ–3dπ < 3dπ–3dπ
- 2pπ–2pπ > 3dπ–3dπ > 3pπ–3dπ > 3pπ–3pπ

Dipole Moment (μ) :- Measure overall polarity in Molecule

Unit = Debye

1 Debye = 3.3×10^{-30} cb·m

$\boxed{\mu = q \times d}$ μ = 0 (Non-polar)
  μ ≠ 0 (Polar)

- vector quantity
- directed towards more EN elements. (More e^- density)
- direction of L.P dipole moment is always away from the molecule.

for All symmetrical structures

AB_2L_0 to AB_7L_0	Non-polar
AB_4L_2, AB_2L_3	Always μ = 0
	S.A must be same.

⇒ Compare dipole Moment

① $CO_2 < SO_2$
 AB_2L_0 AB_2L_1
 μ = 0 μ ≠ 0

② $SO_2 > SO_3$
 AB_2L_1 AB_3L_0
 μ ≠ 0 μ = 0

③ $CCl_4 < CH_2F_2$
 AB_4L_0 AB_4L_0
 μ = 0 μ ≠ 0
 ∵ SA diff.

④ $XeF_2 = XeF_4$
 AB_2L_3 AB_4L_2
 μ = 0 μ = 0

⑤ $NH_3 > NF_3$
 AB_3L_1 AB_3L_1
 μ ≠ 0 μ ≠ 0

⑥ $H_2O > OF_2$
 AB_2L_2 AB_2L_2
 μ ≠ 0 μ ≠ 0
 ↙ same

↳ In NH_3, L.P dipole moment & Bond pair dipole moment are in same direction. ∴ μ ↑

⇒ Examples based on 'q' (ΔEN ↑ |q| ↑ μ ↑)

① HF > HCl > HBr > HI

② HF > NH_3 > PH_3

③ $H_2O > NH_3 > PH_3$ ④ $H_2O > H_2S$

⇒ Some data based examples

① $CH_3Cl > CH_3F > CH_3Br > CH_3I$

② $CH_3Cl > CH_2Cl_2 > CHCl_3 > CCl_4$

③ $H_2O > HF$ (Acc. to NCERT)

⇒ Calculation of Ionic character

$$\boxed{\% IC = \frac{\mu_{Real}}{\mu_{Imaginary}} \times 100} \quad \mu = q \times d$$

Q→ If $\mu_{HCl} = 1.02D \rightarrow 1.02 \times 3.3 \times 10^{-30}$ C.m

$d_{HCl} = 1.27 A^{\circ} \rightarrow 1.27 \times 10^{-10}$ m.

Find % IC in HCl?

Ans:- $\% IC = \frac{1.02 \times 3.3 \times 10^{-30}}{\underset{(q)}{1.6 \times 10^{-19}} \times \underset{(d)}{1.27 \times 10^{-10}}} \times 100$

Ionic Bond | Electrovalent Bond

- formed by complete transfer of electron
- Electrostatic |coulombic force of attraction b/w oppositely charged ions (q^+ & q^-).

$$\boxed{F = \frac{K q_1 q_2}{r^2}}$$

- Generally formed b/w Metals & Non-metals
- Generally Solid, brittle in nature at room temp.
- Do not show isomerism (due to presence of directional Bond)
- Polar & Readily soluble in polar solvent like water.

$\boxed{NH_4Cl}$ → Covalent (NH_3)
coordinate ($\ddot{N}H_3 + H^+$)
Ionic (NH_4^+ & Cl^-)

Hydration :- Break into ions.

↳ Tendency of Ionic compounds.

$\boxed{\text{Degree of Hydration} \propto \text{Ionic potential or Hydration energy}}$

$$\boxed{\text{Ionic potential}(\phi) = \frac{\text{charge}}{\text{Radius}}}$$

$Li^+ < Be^{2+}$ (ϕ)

Li^+ Ⓐ Na^+ Ⓑ K^+ Ⓒ Rb^+ Ⓓ Cs^+ Ⓔ →

$\phi \downarrow$, Ionic size ↑, (degree of hydration) ↓ (hydrated ion size) ↓ , Ionic Mobility or Conductance ↑

⇒ Ionic compounds have M.P & B.P as compare to covalent compounds.

$\boxed{\text{Ionic compounds}}$

- Exist in form of Ionic lattice.
- Do not exist in Molecular form due to strong electrostatic force of attraction.

★ Electrical conductivity of I.C

	Ion	Mobility	Conductance
Solid	✓	✗	✗
Liq (Molten)	✓	✓	✓
~~Aqueous~~ Aquous (In form of hydrated ion)	✓	✓	✓

NOTE → Metals are always conductor of electricity due to presence of free e^-.

$\boxed{\text{Covalent compounds (Molecular Solid)}}$

- Exist in form of Molecules & weak vanderwaal force present b/w molecules.

★ Electrical conductivity

	Ions	Mobility	Conductance
Solid	✗	✗	✗
Liq (Molten)	✗	✗	✗
Aquous	✓	✓	✓

★ To form stable compound :-

Released Energy (Profit) > Absorbed energy (Investment)

EXO > ENDO

E.A, L.E > S.E, I.E, B.D.E

Lattice Energy (U) :- Energy req. to break 1 Mole Ionic solid into its gaseous constituents ions.

$\boxed{U \propto \frac{|q_1||q_2|}{r}}$ $|q|\uparrow, U\uparrow$

If charge is same

$\boxed{U \propto \frac{1}{r}}$ $r\uparrow U\downarrow$

Eg→
$\overset{+1}{Na}\overset{-1}{Cl} < \overset{+1}{Na}\overset{-1}{F}$
$\overset{+2}{Mg}\overset{-2}{O} < \overset{+3}{Al_2}\overset{-2}{O_3}$
$\overset{+1}{Li_3}\overset{-3}{N} > \overset{+1}{Na_3}\overset{-2}{N}$

NOTE:-

AM → Only 'Li' can form stable nitride

AEM → All can form stable nitride due to high L.E (high charge & small size)

Eg→ Be_3N_2, Mg_3N_2

Fajan's Rule → Development of covalent character in ionic compounds.

→ Cation have more Zeff than Anion.

→ Distortion in e^- cloud of Anion

→ Merging e^- clouds of cation & Anion.

$\boxed{\text{Polarisation}\uparrow, \text{covalent character}\uparrow}$

- Polarising Power ∝ Polarisation (ϕ) (for cations)
- Polarisability ∝ Polarisation (for Anions)

Condition for Max. covalent character

→ $r^+\downarrow$ (Small cation) $Be^{2+} > Mg^{2+} > Ca^{2+} > Sr^{2+}$

→ $r^-\uparrow$ (Large Anion) $F^- < Cl^- < Br^- < I^-$

→ $q^+, q^-\uparrow$ (high charge on ions)

Eg→ $M^+ < M^{2+} < M^{3+}$

$X^- < X^{2-} < X^{3-}$

Important points

$\boxed{\phi \text{ for cation} = \frac{q^+ \text{(charge)}}{r^+ \text{(radius)}}}$ $r^+\uparrow$ $\phi\downarrow$

(i) Na^+, K^+, Rb^+, Cs^+ = Low ϕ

(ii) D-block cations ϕ > S-block cations (size↓, Zeff↑, $\phi\uparrow$)

(iii) $LiCl, BeCl_2, MgCl_2$ soluble in organic solvents like benzene, pyridine and Acetone.

covalent char due to very high ϕ nature

(Non-polar solvent)

(iv) Due to covalent character, M.P of ionic compounds decreases.

$\overset{+2}{Sn}Cl_2 < \overset{+4}{Sn}Cl_4$ (cov. ch.)

$SnCl_2 > SnCl_4$ (M.P)

Q→ Compare covalent character

① $LiF < LiCl < LiBr < LiI$ ($r^-\uparrow$)

② $LiCl > NaCl > KCl > RbCl$ ($r^+\downarrow$)

③ $\overset{+2}{Sn}Cl_2 < \overset{+4}{Sn}Cl_4$ (high charge on ions)

④ $NaCl < CuCl$ (d-block > S-block)

Molecular orbital theory

- Paramagnetic behavior of O_2 is explained by MOT.
- Applied only for homodiatomic molecules (total electrons up to 20).
- The Molecular orbitals like A.O are filled in accordance with the aufbau principle obeying the Pauli's exclusion principle & the Hund's rule.
- Acc to MOT

$$A.O + A.O \rightarrow \begin{cases} \text{ABMO (high energy)} & \text{Antibonding MO} \\ \text{BMO (low energy)} & \text{Bonding M.O} \end{cases}$$

Electronic configuration (IMP)

- Beyond N_2 (1221) [Without S-P mixing] (Normal)

$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \begin{matrix}\pi 2p_x \\ \| \\ \pi 2p_y\end{matrix} < \begin{matrix}\pi^* 2p_x \\ \| \\ \pi^* 2p_y\end{matrix} < \sigma^* 2p_z$

1 2 2 1

- Upto N_2 (2121) [With S-P Mixing] (Abnormal)

" " $< \begin{matrix}\pi 2p_x \\ \| \\ \pi 2p_y\end{matrix} < \sigma 2p_z < \begin{matrix}\pi^* 2p_x \\ \| \\ \pi^* 2p_y\end{matrix} < \sigma^* 2p_z$

2 1 2 1

eg→

① O_2 (1221) → 16 e^-

$\sigma 1s^{2}\ \sigma^* 1s^{2}\ \sigma 2s^{2}\ \sigma^* 2s^{2}\ \sigma 2p_z^{2}\ \pi 2p_x^{2}\ \pi 2p_y^{2}\ \pi^* 2p_x^{1}\ \pi^* 2p_y^{1}\ \sigma^* 2p_z$

$B.O = \frac{10-6}{2} = 2$

2 unpaired e^-

∴ O_2 is paramagnetic

② N_2 (2,14) → $14e^-$

$\sigma 1s^2\ \sigma^*1s^2\ \sigma 2s^2\ \sigma^*2s^2\ \pi 2p_x^2 = \pi 2p_y^2\ \sigma 2p_z^2\ \pi^*2p_x = \pi^*2p_y\ \sigma^*2p_z$

$B.O = \frac{6-0}{2} = 3$

$$\boxed{\text{Bond order} = \frac{BMOe^- - ABMOe^-}{2}}$$

- B.O ↑, stability ↑, B.S ↑, B.L ↓

NOTE → Species having same bond order then ABMO e^- ↑, stability ↓se

Super NOTE :-

Total e^- 1 to 8

odd number → 0.5 B.O

even number < 2, 6 → 1 B.O ; 4, 8 → 0 B.O

Total e^- 14 → 3 B.O

8	9	10	11	12	13	$\boxed{14}$	15	16	17	18	19	20
0	0.5	1	1.5	2	2.5	$\boxed{3}$	2.5	2	1.5	1	0.5	0

← 1 e^- ↓se in BMO, B.O ↓se by 0.5 | 1 e^- ↑se in ABMO → B.O ↓se by 0.5

Magnetic Behavior

Count total e^- → odd (Paramagnetic) ; → Even (Diamagnetic)

except → B_2, O_2, S_2 (Paramagnetic)

LCAO (linear combination of A.O)

(formation of M.O) INA = 2

S–S (σ) → $\sigma 1s, \sigma^*1s, \sigma 2s, \sigma^*2s$

P–P (σ) → $\sigma 2p_z, \sigma^*2p_z$

P–P (π) → $\pi 2p_x, \pi 2p_y, \pi^*2p_x, \pi^*2p_y$

NOTE

- All 'σ' MO (BMO & ABMO) are symmetrical along the bond axis.
- All 'π' MO (BMO, ABMO) are unsymmetrical along the bond axis.

⇒ Magnetic Moment → Jitna unpaired e^- utna point something

(n)
1 1.—
2 2.—
3 3.—
4 4.—

$\mu = \sqrt{n(n+2)}$

$= \sqrt{5(5+2)} = \sqrt{35}$

Hydrogen Bonding :-

when hydrogen is linked with high EN element like F/O/N molecule acquire polarity & two adjacent molecules attract each other by H-bond.

$\overset{+\delta}{H} - \overset{-\delta}{X} \cdots\cdots \overset{+\delta}{H} - \overset{-\delta}{X}$ (H-bonding) $\boxed{X = F/O/N}$

- L.P is essential for H-bonding
- H-Bonding is a special case of dipole-dipole interaction.

★ Classification of H-Bonding

InterMolecular H-Bonding	Intramolecular H-bonding
– Present b/w molecules	– Present within molecules
– chelation (ring) may occurs.	– chelation always occur
– More association b/w Molecules	– Less association
– High M.P & B.P	– Low M.P & B.P
– Less volatile (low v.p)	– More volatile
– More viscosity	– Less viscosity

★ Important points about H-bonding

- cellulose can show H-bonding with water.
- Amino acid can show H-bonding with H_2O
- water is liquid at room temp. while H_2S is gas.
- fire due to petrol can not be extinguish by water.
- fire due to Alcohol can be extinguish by water.
- Isomeric ethers are more volatile than Alcohol.
- High B.P & viscosity can be explained by H-bonding.

↳ ortho-Nitrophenol is more volatile than para-nitrophenol.

↳ ortho-salicylaldehyde is liquid at room temp. while para-salicyladehyde is solid.

* Order of B.P & M.P in group 15,16,17 hydrides:-

↳ M.P & B.P of group 15,16,17 hydrides is mainly depends on weak vanderwaal force (w.vf ∝ M.W) but M.P & B.P of (NH_3, H_2O & HF) is higher than expectation in their respective group due to presence of H-bonding.

(1>2>3>4)

B.P (2,1,1)

NH_3 (2)	H_2O (1)	HF (1)
PH_3 (4)	H_2S (4)	HCl (4)
AsH_3 (3)	H_2Se (3)	HBr (3)
SbH_3 (1)	H_2Te (2)	HI (2)

M.P (1,1,2)

NH_3 (1)	H_2O (1)	HF (2)
PH_3 (4)	H_2S (4)	HCl (4)
AsH_3 (3)	H_2Se (3)	HBr (3)
SbH_3 (2)	H_2Te (2)	HI (1)

$BiH_3 > SbH_3 > NH_3 > AsH_3 > PH_3$

B.P → $H_2O > HF > NH_3$

M.P → $H_2O > NH_3 > HF$

H-bonding strength = $HF > H_2O > NH_3$

($\Delta EN\uparrow$, $|q|\uparrow$, H-bond↑)

H-bonding in ice

↳ 3d H-bonding $\quad \uparrow d = \frac{M}{V\downarrow}$

↳ open cage like structure (V↑ D↓)

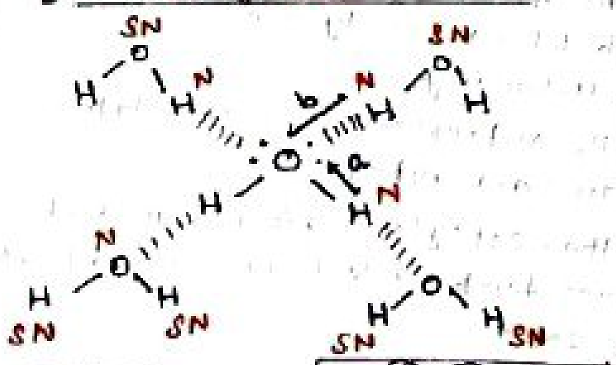
↳ On heating ice contracts (Melts)

V↓ density↑

∴ ice floats on water ← ∴ $d_{water} > d_{ice}$

↳ 'O' atom is surrounded by '4' equidistance 'O' atoms.

[H-bonding from 4 sites]



$\boxed{b > a}$

H-bond B.L — Covalent B.L

	(N)	(SN)	Total
O	2	2	4
H	2	4	6

$H_2O(s)$ → Open cage like structure

$H_3BO_3(s)$ → 2-D Sheet like

$HF(l)$ → zig-zag

$KHCO_3(s)$ → Anionic dimer

$NaHCO_3(s)$ → Anionic polymer

↳ Max^m no. of H-bonds by H_2O in ice is 4.

H-bonding in HF Molecules

↳ zig-zag structure

Dipole Based forces

↳ Interparticle forces ∝ M.W

Ions involved

(1) Ion-dipole (ID)

↳ Present b/w Ions & Polars (Na^+, H_2O)

(2) Ion-induced dipole (IID)

↳ Present b/w ions & Non-polar (Na^+, Br_2)

Ions donot involved (Vanderwaal forces)

(A) Dipole-Dipole (keesom force) (DD)

↳ Present b/w polar Molecules (HCl & HCl)

(B) Dipole-induced dipole (Debye force)

↳ present b/w polar & Non-polar Molecules. [H_2O, Xe & Br_2, H_2O]

(C) LDF or IDID [London dispersion force or instantaneous dipole-induced dipole]

↳ present b/w non-polar molecules.

[H_2, H_2 $\quad F_2, F_2$
Xe, Xe $\quad Cl_2, Cl_2$]

Strength 1 > A > 2 > B > C

ID DD IID DID LDF or IDID

Interaction force (3,4,5,7,7) $\quad F \propto \frac{1}{r^3} \quad \propto \frac{1}{r^4} \quad \propto \frac{1}{r^5} \quad \propto \frac{1}{r^7} \quad \propto \frac{1}{r^7}$

Interaction energy (2,3,4,6,6) $\quad E \propto \frac{1}{r^2} \quad \propto \frac{1}{r^3} \quad \propto \frac{1}{r^4} \quad \propto \frac{1}{r^6} \quad \propto \frac{1}{r^6}$

T to B ↓ M.W↑, VWf↑, London force↑